Title: Evaluating Quantum Phase Estimation Routines Under Realistic Constraints: Implications for Quantum Chemistry Simulation

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Abstract:

Magic state overhead reduction remains a critical challenge for hybrid quantum-classical optimization, particularly in quantum chemistry simulation. This paper proposes a framework built on quantum phase estimation routines that explicitly accounts for crosstalk reduction in multi-qubit devices. We evaluate the framework on the QASMBench circuit suite and compare it against 5 baseline methods. Experimental results show that the proposed approach improves gate fidelity by 35.0% while keeping logical error rate within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

The rapid maturation of hybrid quantum-classical optimization has changed how researchers and practitioners approach magic state overhead reduction, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Despite considerable progress, existing solutions for magic state overhead reduction tend to degrade when moving from controlled laboratory conditions to production workloads, which motivates a more systematic investigation. A central difficulty is that improvements in circuit success probability frequently come at the expense of approximation ratio, forcing designers to accept trade-offs that are poorly understood. This paper addresses these gaps by studying how quantum phase estimation routines interact with quantum annealing schedules when applied to magic state overhead reduction.

Our work is motivated by the growing demand for quantum chemistry simulation, where reliability and transparency requirements are especially strict. We make three contributions: a unified formulation of magic state overhead reduction, a practical design built on quantum phase estimation routines, and an empirical evaluation on the QASMBench circuit suite. Understanding crosstalk reduction in multi-qubit devices is equally important, because it shapes both the theoretical limits and the operational cost of any proposed approach. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

Novak et al. (2021) introduced an early framework based on quantum phase estimation routines, reporting a 26.9% improvement in circuit success probability over previous baselines. Subsequent work by Hoffmann et al. (2024) extended this line of research to quantum chemistry simulation, although the evaluation was restricted to small-scale settings. A survey by Müller et al. (2012) categorised existing approaches to magic state overhead reduction into three families and highlighted the lack of standard benchmarks. Researchers working on hybrid quantum-classical optimization have explored quantum annealing schedules as an alternative, with Singh et al. (2012) noting that the approach scales poorly beyond 128 components. More recently, Vasquez et al. (2014) combined quantum phase estimation routines with quantum annealing schedules, achieving a 10.2% reduction in overhead on the QASMBench circuit suite.

Several studies have criticised the reliance on synthetic workloads, arguing that results on the QASMBench circuit suite may not generalise to quantum chemistry simulation. In contrast to these efforts, our work treats magic state overhead reduction and crosstalk reduction in multi-qubit devices jointly rather than as independent

concerns. Industry reports corroborate these academic findings, pointing to magic state overhead reduction as a persistent obstacle for quantum chemistry simulation. Despite this body of literature, no prior study has systematically compared quantum phase estimation routines and quantum annealing schedules under identical experimental conditions.

3. Methodology

We formalise magic state overhead reduction as a constrained optimisation task in which the objective is to maximise gate fidelity while keeping circuit success probability within a prescribed budget. The proposed architecture consists of four stages: data acquisition, preprocessing, core processing based on quantum phase estimation routines, and a validation layer that incorporates quantum annealing schedules. All experiments were conducted on the QASMBench circuit suite, which provides 8 labelled instances or configurations covering a broad range of operating conditions. To ensure reproducibility, every experiment was repeated 5 times with independent random seeds, and we report mean values together with 95% confidence intervals.

Hyperparameters were selected through grid search over a validation split, and the final configuration was fixed before any evaluation on held-out data. The core routine applies quantum phase estimation routines to reduce the effect of crosstalk reduction in multi-qubit devices, followed by a calibration step that stabilises circuit success probability. Our implementation is written in Python and C++, and was executed on a workstation with 64 GB of memory and a single mid-range accelerator. We compare against 5 baseline methods, including a conventional approach built on quantum annealing schedules and two recent variants reported in the literature.

The proposed design improves gate fidelity by 35.0% relative to the strongest baseline and reduces gate fidelity by 8.3%, a statistically significant difference ($p < 0.05$). A sensitivity analysis over 16 parameter settings shows that performance remains stable, with variation in approximation ratio below 3.5%. Threats to validity include the limited diversity of the QASMBench circuit suite and the possibility that deployment in quantum chemistry simulation introduces conditions not captured in our setup.

4. Conclusion

This paper investigated magic state overhead reduction through the lens of quantum phase estimation routines, and demonstrated that a carefully designed pipeline can improve gate fidelity by 35.0% on the QASMBench circuit suite. The findings suggest that hybrid quantum-classical optimization will continue to benefit from closer integration of quantum phase estimation routines and quantum annealing schedules, particularly in quantum chemistry simulation. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of crosstalk reduction in multi-qubit devices. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for magic state overhead reduction.